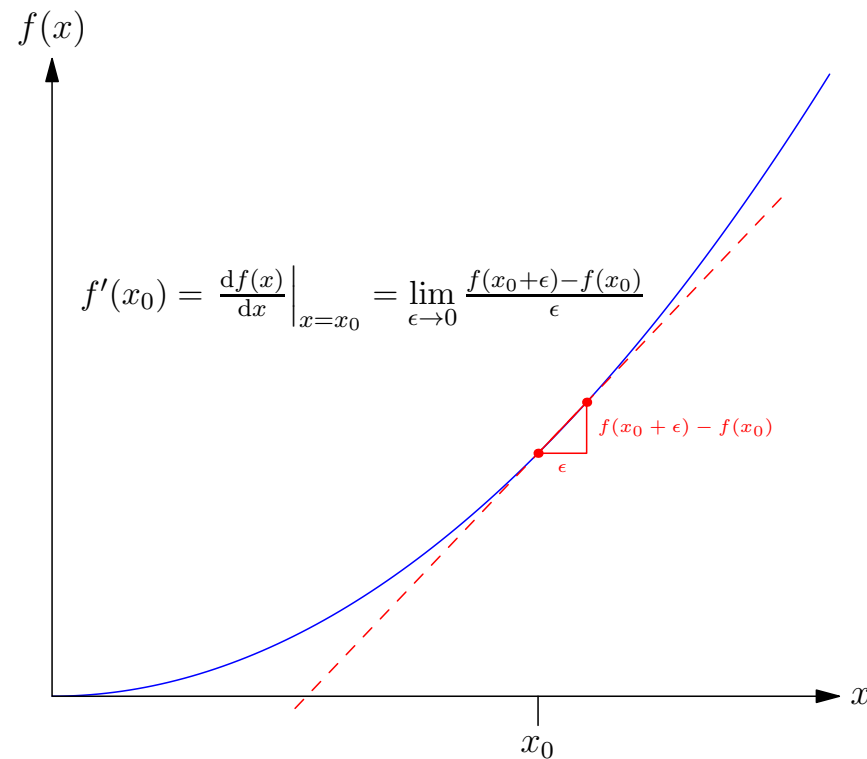


Mathematics of Machine Learning

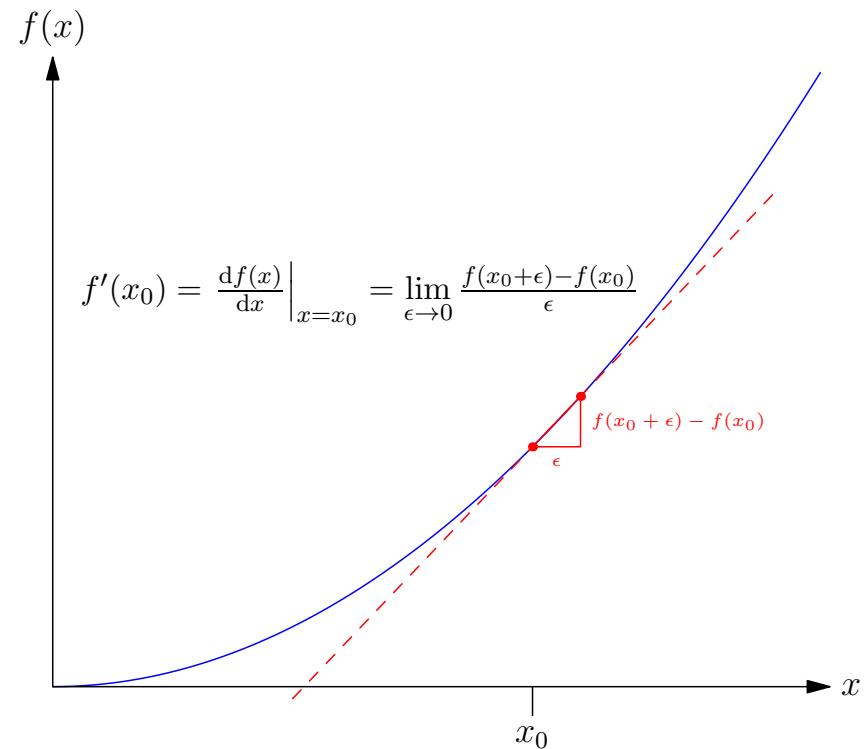
Differential Calculus



Differentiation, product and chain rules, vectors and matrices

Outline

1. **Why Calculus?**
2. Differentiation
3. Vector Calculus
4. Matrix Calculus



Why Calculus?

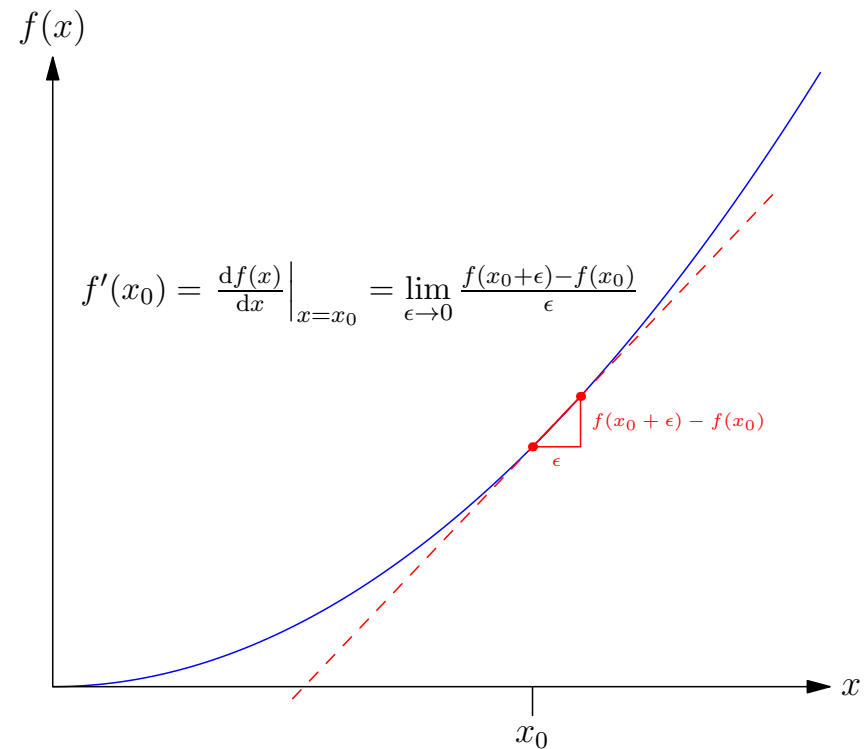
- Calculus is a fundamental tool of mathematical analysis■
- In machine learning differentiation is fundamental tool in optimisation■
- Integration is an essential tool in taking expectations over continuous distributions■
- Both differentiation and integration crop up elsewhere■

Back to Basics

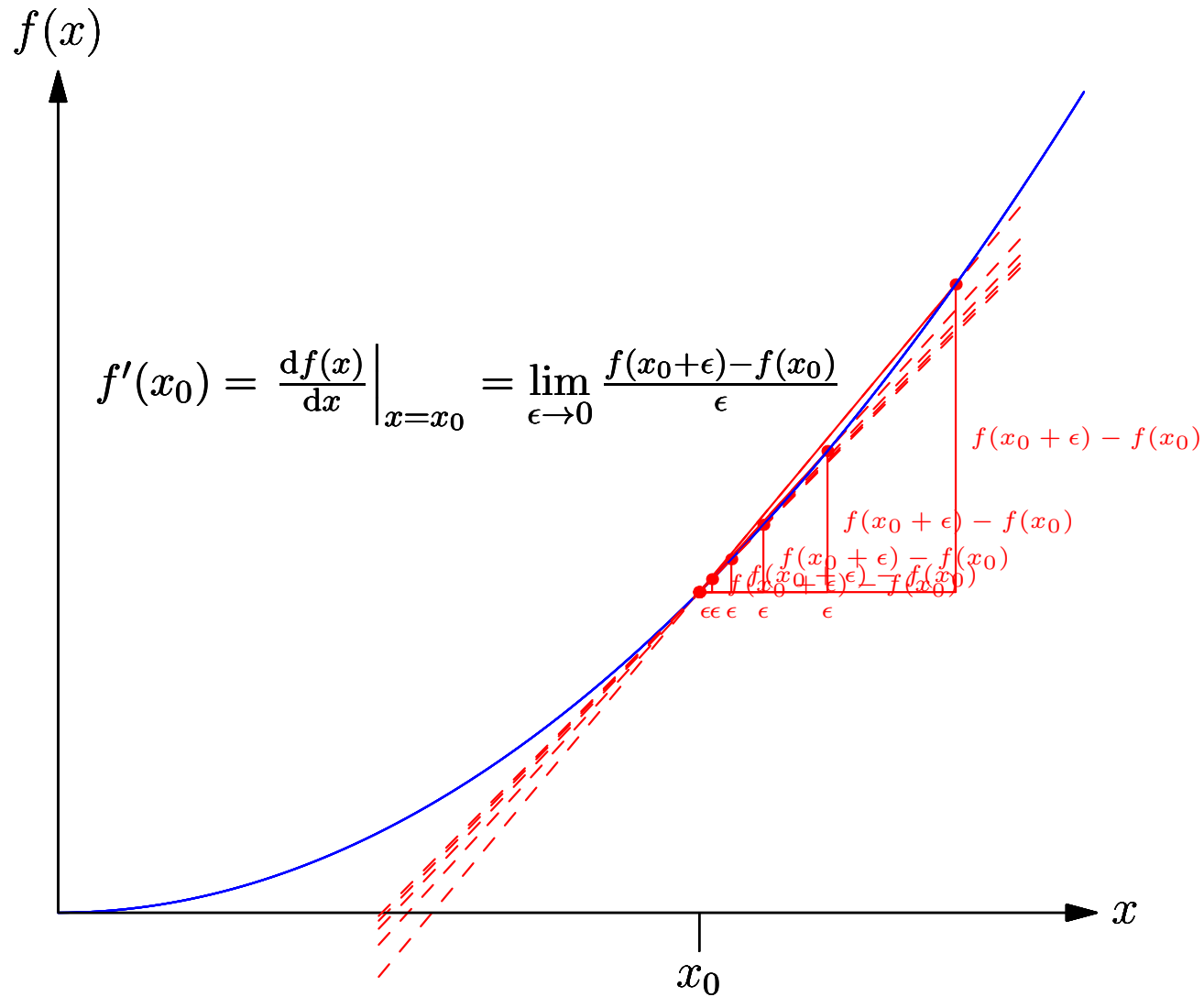
- You have all done A-level maths so should be familiar with the rules of calculus■
- But, it is easy to forget the rules and sometimes we use quite sophisticated tricks■
- Although the sophisticated tricks really speed up calculations, it pays to be able to understand where these tricks come from■

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1. Why Calculus?
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Differentiation



Polynomials

- $f(x) = x^2$

$$\begin{aligned}\frac{dx^2}{dx} &= \lim_{\epsilon \rightarrow 0} \frac{(x + \epsilon)^2 - x^2}{\epsilon} = \lim_{\epsilon \rightarrow 0} \frac{(x^2 + 2\epsilon x + \epsilon^2) - x^2}{\epsilon} \\ &= \lim_{\epsilon \rightarrow 0} 2x + \epsilon = 2x\end{aligned}$$

- $(x + \epsilon)^n = (x + \epsilon)(x + \epsilon) \cdots (x + \epsilon) = x^n + n\epsilon x^{n-1} + O(\epsilon^2)$

$$\frac{dx^n}{dx} = \lim_{\epsilon \rightarrow 0} \frac{(x + \epsilon)^n - x^n}{\epsilon} = \lim_{\epsilon \rightarrow 0} n x^{n-1} + O(\epsilon) = n x^{n-1}$$

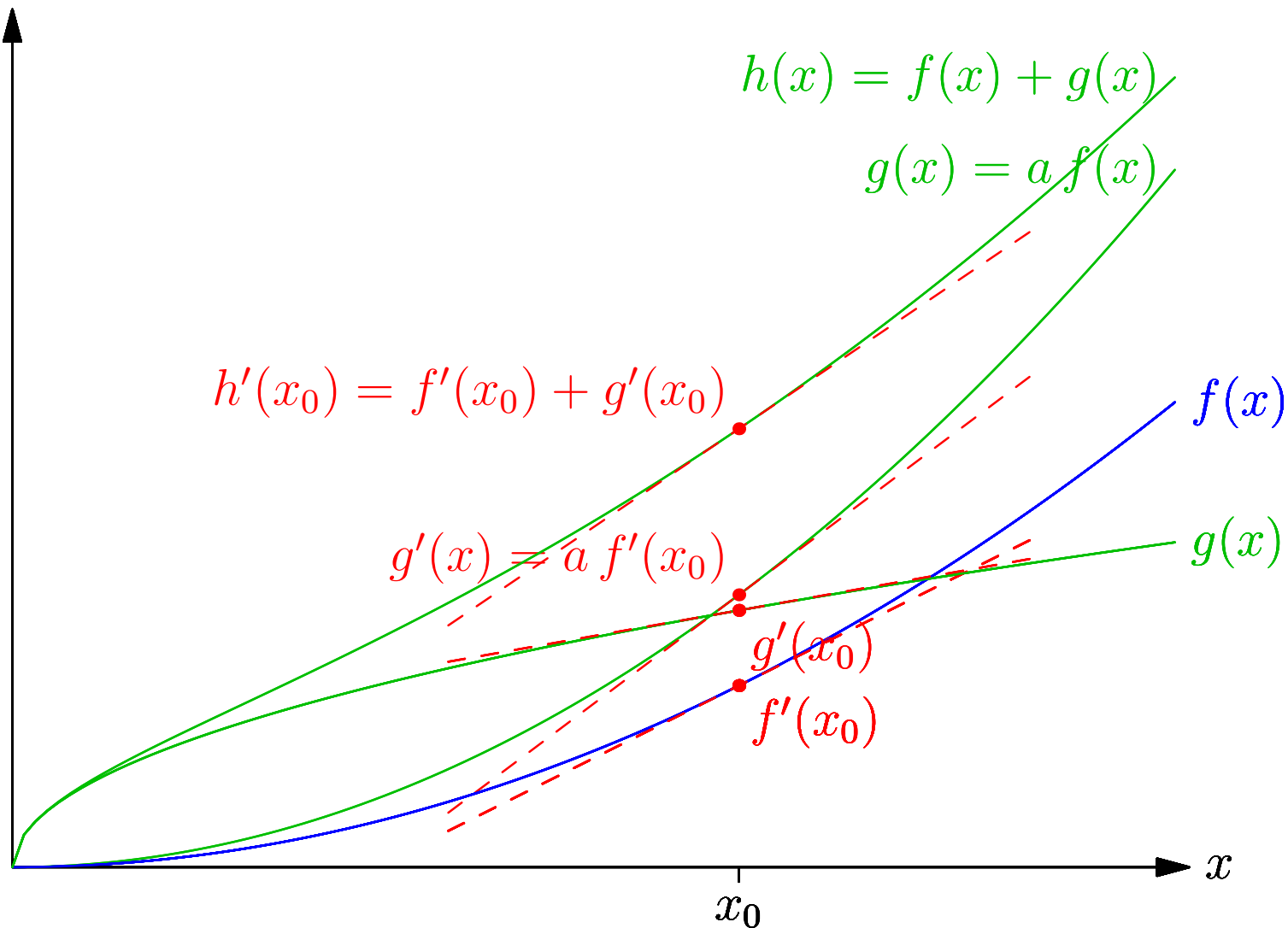
Linearity of derivatives

- Consider the derivative of $a f(x) + b g(x)$

$$\begin{aligned}\frac{d(a f(x) + b g(x))}{dx} &= \lim_{\epsilon \rightarrow 0} \frac{(a f(x + \epsilon) + b g(x + \epsilon)) - (a f(x) + b g(x))}{\epsilon} \\&= \lim_{\epsilon \rightarrow 0} \frac{a(f(x + \epsilon) - f(x)) + b(g(x + \epsilon) - g(x))}{\epsilon} \\&= \lim_{\epsilon \rightarrow 0} a \left(\frac{f(x + \epsilon) - f(x)}{\epsilon} \right) + b \left(\frac{g(x + \epsilon) - g(x)}{\epsilon} \right) \\&= a f'(x) + b g'(x)\end{aligned}$$

- Differentiation is a linear operation!**

Linearity in Pictures



Product Rule

- Recall $f(x + \epsilon) = f(x) + \epsilon f'(x) + O(\epsilon^2)$
- If $h(x) = f(x)g(x)$

$$\begin{aligned} h'(x) &= \lim_{\epsilon \rightarrow 0} \frac{f(x + \epsilon)g(x + \epsilon) - f(x)g(x)}{\epsilon} \\ &= \lim_{\epsilon \rightarrow 0} \frac{(f(x) + \epsilon f'(x) + O(\epsilon^2))(g(x) + \epsilon g'(x) + O(\epsilon^2)) - f(x)g(x)}{\epsilon} \\ &= \lim_{\epsilon \rightarrow 0} \frac{\epsilon(f'(x)g(x) + f(x)g'(x)) + O(\epsilon^2)}{\epsilon} = f'(x)g(x) + f(x)g'(x) \end{aligned}$$

- This is the **product rule**

Chain Rule

- Recall $f(x + \epsilon) = f(x) + \epsilon f'(x) + O(\epsilon^2)$
- Let $h(x) = f(g(x))$ ■
- Then

$$\begin{aligned} h(x + \epsilon) &= f(g(x + \epsilon)) \text{■} = f(g(x) + \epsilon g'(x) + O(\epsilon^2)) \text{■} \\ &= f(g(x)) + \epsilon g'(x) f'(g(x)) + O(\epsilon^2) \text{■} \end{aligned}$$

- Thus

$$h'(x) = \lim_{\epsilon \rightarrow 0} \frac{h(x + \epsilon) - h(x)}{\epsilon} = g'(x) f'(g(x)) \text{■}$$

- This is the famous **chain rule**■ Together with the product rule it means you can differentiate almost everything■

More on chain rules

- We can also write the chain rule as

$$\frac{df(g(x))}{dx} = \frac{df(g)}{dg} \frac{dg(x)}{dx}$$

- Sometimes this is neater or easier to remember

$$\begin{aligned} \frac{de^{\cos(x^2)}}{dx} &= \frac{de^{\cos(x^2)}}{d\cos(x^2)} \frac{d\cos(x^2)}{dx^2} \frac{dx^2}{dx} \\ &= e^{\cos(x^2)} (-\sin(x^2)) 2x \\ &= -2x \sin(x^2) e^{\cos(x^2)} \end{aligned}$$

Inverse functions

- Suppose $g(y) = f^{-1}(y)$ is the inverse of $f(x)$ in the sense that $g(f(x)) = f^{-1}(f(x)) = x$
- Using the chain rule

$$\frac{dg(f(x))}{dx} = f'(x)g'(f(x)) = 1$$

since $g(f(x)) = x$

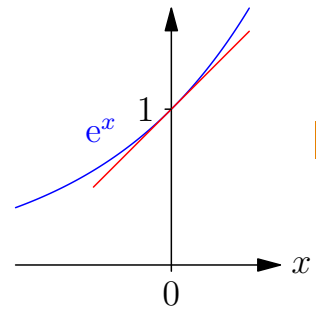
- So $g'(f(x)) = 1/f'(x)$
- Writing $y = f(x)$ so that $x = f^{-1}(y) = g(y)$ we find $g'(y) = 1/f'(g(y))$ that is

$$\frac{dg(y)}{dy} = \frac{1}{f'(g(y))} \qquad \frac{df^{-1}(y)}{dy} = \frac{1}{f'(f^{-1}(y))}$$

Exponentials

- Note that $a^{b+c} = a^b a^c$ (that is we multiply a together $b + c$ times)■

- Now $e^\epsilon \approx (1 + \epsilon)$



- But $e^{x+\epsilon} = e^x e^\epsilon = e^x (1 + \epsilon + O(\epsilon^2))$ ■ $= e^x + \epsilon e^x + O(\epsilon^2)$ ■

$$\frac{de^x}{dx} = \lim_{\epsilon \rightarrow 0} \frac{e^{x+\epsilon} - e^x}{\epsilon} \text{■} = \lim_{\epsilon \rightarrow 0} \frac{\epsilon e^x + O(\epsilon^2)}{\epsilon} = e^x \text{■}$$

Functions of Exponentials

- What about $f(x) = e^{cx}$

$$\frac{de^{cx}}{dx} = \frac{de^{cx}}{dcx} \frac{dcx}{dx} = ce^{cx}$$

- More generally using the chain rule

$$\frac{de^{g(x)}}{dx} = g'(x)e^{g(x)}$$

- Also $a^{bc} = (a^b)^c$ (that is we multiply a together $b \times c$ times)

$$\frac{da^x}{dx} = \frac{d(e^{\ln(a)})^x}{dx} = \frac{de^{\ln(a)x}}{dx} = \ln(a)e^{\ln(a)x} = \ln(a)a^x$$

Natural Logarithms

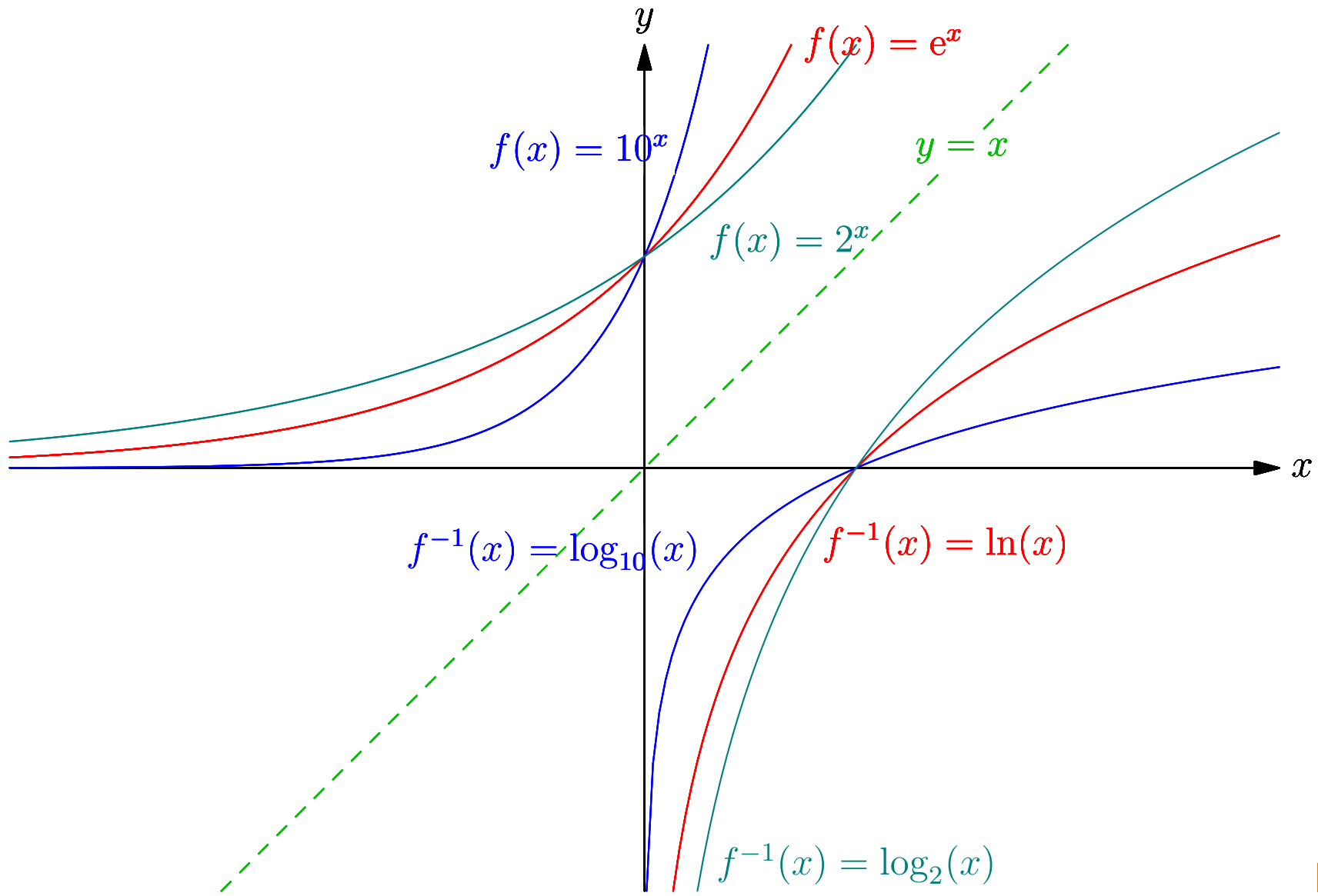
- The natural logarithm is defined as the inverse of e^x

$$\ln(e^x) = x \qquad e^{\ln(y)} = y$$

- Recall that if $g(y) = f^{-1}(y)$ then $g'(y) = 1/f'(g(y))$
- Consider $g(y) = \ln(y)$ and $f(x) = e^x$ (with $f'(x) = e^x$)

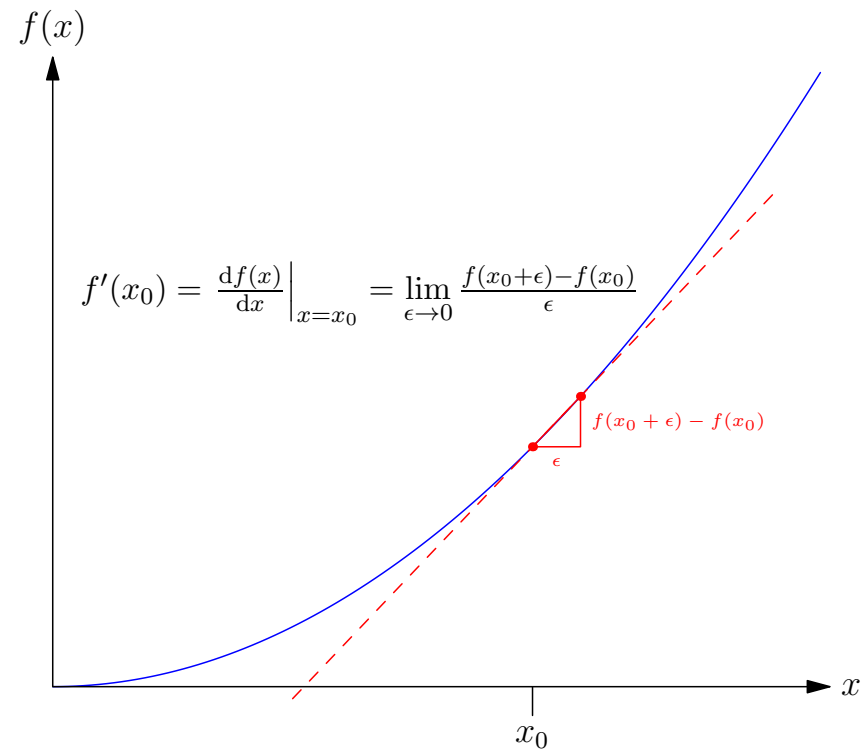
$$\frac{d\ln(y)}{dy} = \frac{1}{e^{\ln(y)}} = \frac{1}{y}$$

Exponentials and Logarithms



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Derivatives in High Dimensions

- When working with functions $f : \mathbb{R}^n \rightarrow \mathbb{R}$ in many dimensions then there will typically be different derivative in different directions■
- To compute the derivative in a direction $\mathbf{u} \in \mathbb{R}^n$ (where $\|\mathbf{u}\| = 1$) at a point $\mathbf{x} \in \mathbb{R}^n$ we use

$$\partial_{\mathbf{u}} F(\mathbf{x}) = \lim_{\epsilon \rightarrow 0} \frac{f(\mathbf{x} + \epsilon \mathbf{u}) - f(\mathbf{x})}{\epsilon} \blacksquare$$

- If $\mathbf{u} = \boldsymbol{\delta}_i = (0, \dots, 0, 1, 0, \dots, 0)^\top$ (i.e. $u_i = 1$) then

$$\frac{\partial f(\mathbf{x})}{\partial x_i} = \lim_{\epsilon \rightarrow 0} \frac{f(\mathbf{x} + \epsilon \boldsymbol{\delta}_i) - f(\mathbf{x})}{\epsilon} \blacksquare$$

Taylor

- If we expand $f(\mathbf{x} + \epsilon \mathbf{u})$ to first order in ϵ

$$f(\mathbf{x} + \epsilon \mathbf{u}) = f(\mathbf{x}) + \epsilon \mathbf{u}^\top \mathbf{g}(\mathbf{x}) + O(\epsilon^2)$$

then $g_i(\mathbf{x}) = \frac{\partial f(\mathbf{x})}{\partial x_i}$ ■

- Recall we defined the vector of first order derivatives of $f(\mathbf{x})$ to be the gradient

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \frac{\partial f(\mathbf{x})}{\partial x_2} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_n} \end{pmatrix} \quad \blacksquare$$

- Thus

$$f(\mathbf{x} + \epsilon \mathbf{u}) = f(\mathbf{x}) + \epsilon \mathbf{u}^\top \nabla f(\mathbf{x}) + O(\epsilon^2) \quad \blacksquare$$

This is the start of the high-dimensional Taylor expansion ■

Computing Gradients 1

- We can compute the gradient by writing out $f(\mathbf{x})$ componentwise and performing the partial derivative with respect to x_i

$$\begin{aligned}\nabla \mathbf{w}^\top \mathbf{M} \mathbf{w} &= \begin{pmatrix} \frac{\partial}{\partial w_1} \\ \frac{\partial}{\partial w_2} \\ \frac{\partial}{\partial w_3} \\ \vdots \end{pmatrix} \sum_{i,j} w_i M_{ij} w_j = \begin{pmatrix} \sum_j M_{1j} w_j + \sum_i w_i M_{i1} \\ \sum_j M_{2j} w_j + \sum_i w_i M_{i2} \\ \sum_j M_{3j} w_j + \sum_i w_i M_{i3} \\ \vdots \end{pmatrix} \\ &= \mathbf{M} \mathbf{w} + \mathbf{M}^\top \mathbf{w} = (\mathbf{M} + \mathbf{M}^\top) \mathbf{w}\end{aligned}$$

- It is tedious to compute these things component-wise, but when you need to understand what is going on then go back to the basics

Computing Gradients 2

- A slicker way is just to expand $f(x + \epsilon u)$ ■

- Consider $f(x) = x^\top M x + a^\top x$

$$\begin{aligned} f(x + \epsilon u) &= (x + \epsilon u)^\top M (x + \epsilon u) + a^\top (x + \epsilon u) \blacksquare \\ &= f(x) + \epsilon (u^\top M x + x^\top M u + a^\top u) + O(\epsilon^2) \blacksquare \\ &= f(x) + \epsilon u^\top (M x + M^\top x + a) + O(\epsilon^2) \end{aligned}$$

using $x^\top M u = u^\top M^\top x$ and $a^\top u = u^\top a$ ■

- But $f(x + \epsilon u) = f(x) + \epsilon u^\top \nabla f(x) + O(\epsilon^2)$ so

$$\nabla f(x) = M x + M^\top x + a \blacksquare$$

Function Composition

- The gradient tells us about the change in a function value when we perturb an input $\mathbf{x}$ by a small amount in the directions $\mathbf{u}$

$$f(\mathbf{x} + \epsilon \mathbf{u}) = f(\mathbf{x}) + \epsilon \mathbf{u}^\top \nabla f(\mathbf{x}) + O(\epsilon^2) \blacksquare$$

- What happens when we compose functions: $h(\mathbf{x}) = f(\mathbf{g}(\mathbf{x}))$ $\blacksquare$
Note that,

$$\mathbf{x} \in \mathbb{R}^n \quad \mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^m \quad f : \mathbb{R}^m \rightarrow \mathbb{R} \quad h : \mathbb{R}^n \rightarrow \mathbb{R} \blacksquare$$

- The vector valued function $\mathbf{g}(\mathbf{x})$ can be seen as m independent

$$\text{function } \mathbf{g}(\mathbf{x}) = \begin{pmatrix} g_1(\mathbf{x}) \\ g_2(\mathbf{x}) \\ \vdots \\ g_m(\mathbf{x}) \end{pmatrix} \blacksquare$$

Jacobians

- Perturbing $\mathbf{x}$ by $\epsilon \mathbf{u}$

$$g_i(\mathbf{x} + \epsilon \mathbf{u}) = g_i(\mathbf{x}) + \epsilon \mathbf{u}^\top \nabla g_i(\mathbf{x}) + O(\epsilon^2)$$

so that

$$\mathbf{g}(\mathbf{x} + \epsilon \mathbf{u}) = \mathbf{g}(\mathbf{x}) + \epsilon \mathbf{J} \mathbf{u} + O(\epsilon^2)$$

where $\mathbf{J}$ is an $m \times n$ matrix with components $J_{ij} = \frac{\partial g_i(\mathbf{x})}{\partial x_j}$

- Note that $\mathbf{J} \mathbf{u}$ is a column vector of size m
- Ignoring terms of order ϵ^2

$$\begin{aligned} h(\mathbf{x} + \epsilon \mathbf{u}) &= f(\mathbf{g}(\mathbf{x} + \epsilon \mathbf{u})) \\ &= f(\mathbf{g}(\mathbf{x}) + \epsilon \mathbf{J} \mathbf{u}) \\ &= f(\mathbf{g}(\mathbf{x})) + \epsilon \mathbf{u}^\top \mathbf{J}^\top \nabla_{\mathbf{g}} f(\mathbf{g}(\mathbf{x})) \end{aligned}$$

Chain Rule in High Dimensions

- Using $h(\mathbf{x}) = f(\mathbf{g}(\mathbf{x}))$

$$h(\mathbf{x} + \epsilon \mathbf{u}) = h(\mathbf{x}) + \epsilon \mathbf{u}^\top \mathbf{J}^\top \nabla_{\mathbf{g}} f(\mathbf{g}(\mathbf{x})) \blacksquare$$

- We get that

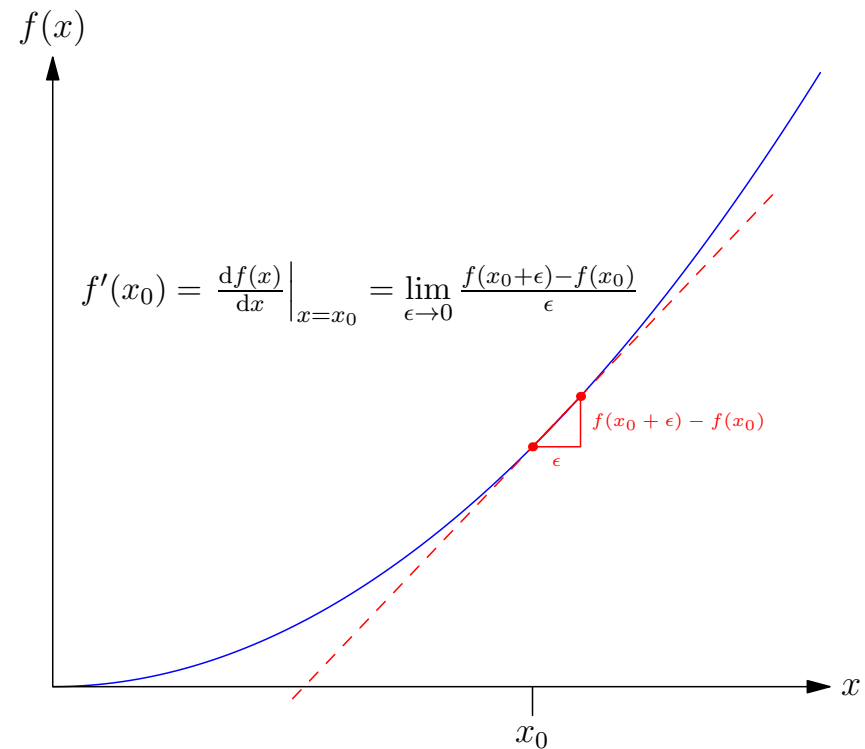
$$\nabla_{\mathbf{x}} h(\mathbf{x}) = \nabla_{\mathbf{x}} f(\mathbf{g}(\mathbf{x})) = \mathbf{J}^\top \nabla_{\mathbf{g}} f(\mathbf{g}(\mathbf{x})) \blacksquare$$

- Componentwise

$$\frac{\partial h(\mathbf{x})}{\partial x_i} = \frac{\partial f(\mathbf{g}(\mathbf{x}))}{\partial x_i} \blacksquare = \sum_{j=1}^m \frac{\partial g_j(\mathbf{x})}{\partial x_i} \frac{\partial f(\mathbf{g}(\mathbf{x}))}{\partial g_j} \blacksquare$$

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Differentiating Matrices

- Often we have loss functions with respect to a matrix $\mathbf{W}$, e.g.

$$L(\mathbf{W}) = (\mathbf{a}^\top \mathbf{W} \mathbf{b} - c)^2$$

- We might want to find the minimum with respect to $\mathbf{W}$
- This occurs at a point $\mathbf{W}^*$ where $L(\mathbf{W})$ does not increase as we change $\mathbf{W}$ in any way
- That is, we seek a $\mathbf{W}^*$ such that, for any matrices $\mathbf{U}$

$$L(\mathbf{W}^* + \epsilon \mathbf{U}) - L(\mathbf{W}^*) = O(\epsilon^2)$$

Generalised Gradient

- We can generalise the idea of gradient to matrices

$$\frac{\partial L(\mathbf{W})}{\partial \mathbf{W}} = \begin{pmatrix} \frac{\partial L(\mathbf{W})}{\partial W_{11}} & \frac{\partial L(\mathbf{W})}{\partial W_{12}} & \cdots & \frac{\partial L(\mathbf{W})}{\partial W_{1m}} \\ \frac{\partial L(\mathbf{W})}{\partial W_{21}} & \frac{\partial L(\mathbf{W})}{\partial W_{22}} & \cdots & \frac{\partial L(\mathbf{W})}{\partial W_{2m}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial L(\mathbf{W})}{\partial W_{n1}} & \frac{\partial L(\mathbf{W})}{\partial W_{n2}} & \cdots & \frac{\partial L(\mathbf{W})}{\partial W_{nm}} \end{pmatrix} \quad \blacksquare$$

- From an identical argument we used for vectors

$$L(\mathbf{W} + \epsilon \mathbf{U}) = L(\mathbf{W}) + \epsilon \text{tr} \mathbf{U}^\top \frac{\partial L(\mathbf{W})}{\partial \mathbf{W}} + O(\epsilon^2) \quad \blacksquare$$

where

$$\text{tr} \mathbf{U}^\top \mathbf{G} = \sum_i [\mathbf{U}^\top \mathbf{G}]_{ii} = \sum_{ij} U_{ji} G_{ji} = \sum_{ij} U_{ij} G_{ij} = \langle \mathbf{U}, \mathbf{G} \rangle \quad \blacksquare$$

Example

- Suppose

$$L(\mathbf{W}) = (\mathbf{a}^\top \mathbf{W} \mathbf{b} - c)^2$$

then

$$\begin{aligned} L(\mathbf{W} + \epsilon \mathbf{U}) &= (\mathbf{a}^\top (\mathbf{W} + \epsilon \mathbf{U}) \mathbf{b} - c)^2 = (\mathbf{a}^\top \mathbf{W} \mathbf{b} + \epsilon \mathbf{a}^\top \mathbf{U} \mathbf{b} - c)^2 \\ &= L(\mathbf{W}) + 2\epsilon (\mathbf{a}^\top \mathbf{W} \mathbf{b} - c) (\mathbf{a}^\top \mathbf{U} \mathbf{b}) + O(\epsilon^2) \end{aligned}$$

- Now

$$\mathbf{a}^\top \mathbf{U} \mathbf{b} = \sum_{ij} a_i U_{ij} b_j = \sum_{ij} U_{ji} a_j b_i = \text{tr} \mathbf{U}^\top \mathbf{a} \mathbf{b}^\top$$

$$\text{Thus } \frac{\partial L(\mathbf{W})}{\partial \mathbf{W}} = 2 (\mathbf{a}^\top \mathbf{W} \mathbf{b} - c) \mathbf{a} \mathbf{b}^\top$$

Traces

- The trace of a matrix is the sum of its diagonal elements

$$\text{tr} \mathbf{A} = \text{tr} \mathbf{A}^T = \sum_i A_{ii}$$

- Clearly $\text{tr} c\mathbf{A} = c \text{tr} \mathbf{A}$
- Also $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr} \mathbf{A} + \text{tr} \mathbf{B}$
- We note that

$$\text{tr} \mathbf{A} \mathbf{B} = \sum_{i,j} A_{ij} B_{ji} = \sum_{i,j} B_{ij} A_{ji} = \text{tr} \mathbf{B} \mathbf{A}$$

- It follows that

$$\text{tr} \mathbf{A} \mathbf{B} \mathbf{C} \mathbf{D} = \text{tr} \mathbf{D} \mathbf{A} \mathbf{B} \mathbf{C} = \text{tr} \mathbf{C} \mathbf{D} \mathbf{A} \mathbf{B} = \text{tr} \mathbf{B} \mathbf{C} \mathbf{D} \mathbf{A}$$

Quick Matrix Differentiation

- Let

$$\partial_{\mathbf{U}} f(\mathbf{X}) = \lim_{\epsilon \rightarrow 0} \frac{f(\mathbf{X} + \epsilon \mathbf{U}) - f(\mathbf{X})}{\epsilon} = \text{tr } \mathbf{U}^T \frac{\partial f(\mathbf{X})}{\partial \mathbf{X}}$$

- E.g.

$$\begin{aligned} \partial_{\mathbf{U}} \text{tr } \mathbf{A} \mathbf{X} \mathbf{B} &= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (\text{tr } \mathbf{A} (\mathbf{X} + \epsilon \mathbf{U}) \mathbf{B} - \text{tr } \mathbf{A} \mathbf{X} \mathbf{B}) \\ &= \text{tr } \mathbf{A} \mathbf{U} \mathbf{B} = \text{tr } \mathbf{B}^T \mathbf{U}^T \mathbf{A}^T = \text{tr } \mathbf{U}^T \mathbf{A}^T \mathbf{B}^T \end{aligned}$$

thus

$$\frac{\partial \text{tr } \mathbf{A} \mathbf{X} \mathbf{B}}{\partial \mathbf{X}} = \mathbf{A}^T \mathbf{B}^T$$

Summary

- With care you can differentiate most expressions■
- The chain and product rule are incredibly powerful tools■
- We can generalise differentiation to vectors and matrices■
- There are a number of surprisingly useful results■ see **The Matrix Cookbook**■
- When we look at **integration** it gets harder■